

NEXATECH

Project Alpha — Deployment Guide

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Purpose

This runbook describes the standard deployment procedure for Project Alpha and the rollback process used when a release causes unacceptable service degradation.

Pre-Deployment Checks

The pull request must be approved, automated tests must pass, dependency and security checks must pass, and the Docker image must be successfully built.

Deployment Sequence

1. Merge the approved pull request. 2. Build and scan the container image. 3. Push the image to the approved registry. 4. Deploy to staging. 5. Run smoke tests. 6. Request production approval. 7. Deploy to production. 8. Monitor service health.

Monitoring

During deployment, monitor HTTP error rate, latency, CPU, memory, PostgreSQL connections, Redis health, queue depth, and agent/tool failures.

Rollback Trigger

If the application error rate exceeds 5 percent for five consecutive minutes, or a critical customer workflow fails, stop the rollout and evaluate rollback.

Rollback Procedure

Stop the deployment, notify the incident channel, deploy the last known-good image, verify health checks, confirm recovery, and create or update an incident record.

Database Changes

Database migrations must be backward compatible where practical. Destructive schema changes require explicit review and a tested recovery plan.

Emergency Changes

Emergency changes may use an expedited process when necessary to restore service. The change must be documented and reviewed after stabilization.

Related Documents

See DOC-SEC-002 Incident Response Procedure and DOC-ENG-002 System Architecture.

This document is synthetic data created for development, RAG, agent orchestration, and evaluation testing.